

Mathematics A

Unit 2 • Chapter OL-T42 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 14 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T42 • Expertise Q16+ • 3 items • 14 marks

Chapter	Items	Marks	Starts
01. OL-T42 Circle Theorems	3	14	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circle Theorems

Unit 2 • Expertise • 14 marks • 3 item(s)

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q22 demand

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

4MA1-2H Q16 M01, M02, M03, M04

5 marks

16 A, B, C and D are points on a circle, centre O

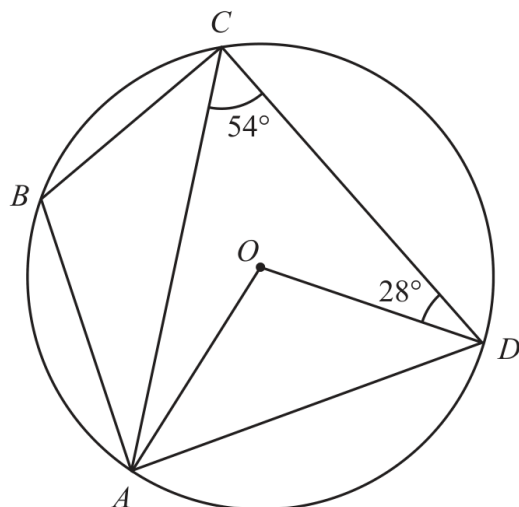


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

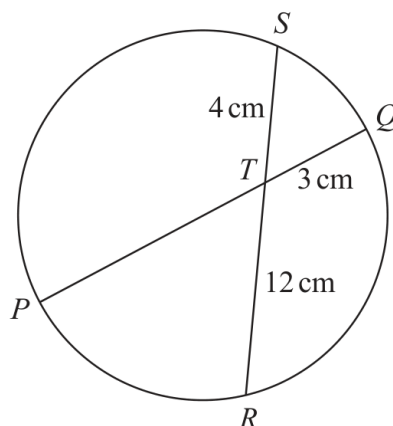
Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q19

3 marks

19

Diagram **NOT**
accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3 \text{ cm} \quad ST = 4 \text{ cm} \quad TR = 12 \text{ cm}$$

Calculate the radius of the circle.

..... cm

Continue your method**3 marks**

WORKING SPACE